Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [12EC82]

EIGHTH SEMESTER B.E UNIVERSITY EXAMINATION, JUNE - 2016

LOW POWER VLSI DESIGN

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1.	a.	With the help of MIS structure, explain the changes occurring at the intrinsic level by applying different voltages.	10
	b.	Explain body effect in MOS devices for low power.	04
2.	a.	Derive the expression for dynamic and short-circuit power dissipations in CMOS circuits.	10
	b.	Explain overlap capacitance of a MOS field-effect transistor.	04
3.	a.	Implement a IIR filter with word parallel or serial data flow graph for the equation. $S_1(t+1)=C_1 S_1(t)+C_3 S_2(t)+K u(t)$, and briefly explain.	08
	b.	Explain how pipelined implementation helps in area and power dissipation.	06
4.	a.	Let $F=\{f_1,f_2\}$ be a two output circuit given by $f_1=ad+bed+ae$ and $f_2=a+bc+dh+eh$. Using power dissipation driven multilevel logic optimization technique, find the total area and the number of transistors required to implement the function $F=\{f_1,f_2\}$ for the kernel intersections $G=\{a+bc,d+e\}$.	10
	b.	Implement the function x^3+Ax^2+Bx+c, and explain the way in which power can be optimized.	04
5.	a.	Explain different types of leakage mechanism in short-channel transistors.	10
	b.	Explain reverse V_{gs} technique in low-voltage circuit design.	04
6.	a.	Explain the following related to V_{th} exploitation to achieve low power. i. SATS ii. MTCMOS iii. Multiple threshold based on critical path.	10
	b.	Explain the importance of multiple supply voltages as applied to low power.	04

7. a. Derive the energy dissipation in transistor channel using a Re model and hence comment on the power consumption related to high and low frequencies. **10**

b. Explain energy recovery address decoder. **04**

8. a. Explain different levels of software power estimation. **10**

b. Explain minimizing memory Access costs for low power. **04**